Uniform Circular Motion: Follow-Along Notes

UNIT 2: CIRCULAR MOTION • DAY 51 LECTURE NOTES (SLIDES 1–50)

Name: _____ **Period:** _____ **Date:** _____

Warm-Up Review: Units of Force

Slides 2 & 18

☐ **IDENTIFY** **Review Question:** What are the units for each of the following forces?

- Force of a **push**: _____
- Force of **friction**: _____
- Force of a **pull**: _____
- Force of **gravity**: _____

Part 1: Circular Movement — Rotation vs. Revolution

Slides 3–12

Circular Movement (Slide 3)

- **Axis:** A straight _____ around which an object rotates.
- **Rotation:** The motion of an object as it turns about its _____ .

ROTATION (Slides 4–5)

- Spinning around _____ .

What else rotates? (Slide 9):

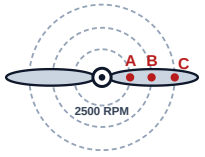
1. _____
2. _____

REVOLUTION (Slides 5–6)

- Moving around _____ .

What else revolves? (Slide 11):

1. _____
2. _____



Propeller (Slides 7–14)

- **Slide 6 (Earth):** The Earth is _____ as it _____ around the sun!
- **Slides 7–10 (Propeller):** The whole propeller _____ , while Points **A**, **B**, and **C** _____ around the axis of the propeller.

☐ **Complete the Sentence (Slide 12):**

_____ , _____ , and _____ are all objects that **rotate** on their axis; while _____ , _____ , and _____ **revolve** around the axis of another object.

Part 2: Rotational Speed vs. Tangential Speed

Slides 13–17

☐ **PREDICT** **Slide 13:** Which point on the propeller is moving the fastest? **Point A** **Point B** **Point C**

☐ **EXPLAIN** **Slide 14:** Because Point _____ travels a _____ distance, in the _____ amount of time, Point _____ is traveling the fastest.

Rotational Speed (Slides 15–17)

- Definition: The number of _____ per second or minute.
- **R.P.M.** = _____
- *Examples:* Dremel Tool (27,000 RPM) & Car **Tachometer**.

Tangential Speed (Slide 15)

- Notes: _____

_____

- On the propeller, all points have the same **RPM** (2500 RPM), but Point _____ has the greatest **tangential speed**.

Part 3: Roller Coaster Loops & Centripetal Force

Slides 19–24

- **Slides 19–22 Question:** When you ride a roller coaster through a loop, why don't you fall out of your seat?

Answer: _____

⚠️ **IMPORTANT!!! (SLIDE 23)**

_____ **Force**

NOT _____ **!**

Turn over for Page 2 (Slides 25–50) →

Part 4: Newton's Second Law & Two Ways to Accelerate

Slides 25–33

Newton's Second Law (Slides 25 & 33): Force = _____ × _____ $F = m\,a$

☐ IDENTIFY **Slide 26:** What are **two ways** to make a mass accelerate?

Way #1 (Slide 27): Change its _____ .
(Up **+** or Down **−**)

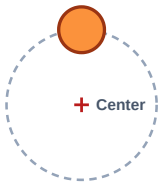
Way #2 (Slide 28): Change its _____ !
Notes: _____

- **Slides 29–31 (The Orange Ball):** The orange ball moving in a circle is changing _____ , so its **velocity** is also _____ .
- **Slide 32 Rule:** Anytime the _____ or _____ changes, the object **IS** _____ !

Part 5: Velocity, Force & Acceleration Vectors

Slides 34–39

DRAW & LABEL THE 3 VECTORS:



Draw Velocity, Force & Acceleration arrows

☐ SKETCH **Slide 35 — Velocity Vector (v):** Draw the velocity arrow on the orange ball at left. Which way does it point? _____

☐ SKETCH **Slides 36–37 — Force Vector (F):** Which way does the force vector point?
_____

☐ SKETCH **Slides 38–39 — Acceleration Vector (a):** Which way does the acceleration vector point?
_____

Part 6: Three Things — Speed, Acceleration & Centripetal Force

Slides 40–50

Slides 40–42 ("3 Things"):	1. How do we determine <u>speed</u>?	2. How do we determine <u>acceleration</u>?	3. How do we calculate <u>force</u>?

1. First: Speed (Slides 43–44)

Slide 44: "What is the speed of this 1 kg orange?"

- How much **distance** does it go in 1 circle? *How far?* = _____
- How much **time** does it take? = _____

Circular Speed Formula: $v = \frac{\square}{\square}$

SLIDE 44 DEMO / SPEED CALCULATION (1 KG ORANGE):

v = _____ m/s

2. Now: Acceleration (Slides 46–48)

How is acceleration determined at constant speed?

a **becomes** $\frac{\square}{\square}$

- v = _____ (units = _____)
- r = _____ (units = _____)

3. Centripetal Force (Slides 47, 49–50)

Substitute circular acceleration into $F = m\,a$:

$F = m\,a \rightarrow F = m\,\frac{\square}{\square}$

☐ **Slide 50: Find this Centripetal Force formula on your Reference Sheet!**